

HUANXUAN LIAO

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🎓 EDUCATION

Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing 2023 – Now

PhD in Computer Science. In NLP, Supervisor: [Shizhu He](#) and [Kang Liu](#)

North China Electric Power University (NCEPU), Beijing 2019 – 2023

Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Ranking 1/13

💡 RESEARCH INTERESTS

- **Long-Context & Multimodal Efficiency** – make long-sequence / multimodal inference practical via compression and input-side budget allocation:
 - Survey & community: [Awesome-LLM-Long-Context-Modeling](#) (GitHub Stars > 2.2K); LCLM survey (core contributor)
 - Context / KV compression: [HyCo2](#) (arXiv 2025), [Spark](#) (AAAI 2026)
 - Parametric RAG: [DyPRAG](#) (arXiv 2025)
 - Multimodal efficiency: [ResAdapt](#) (arXiv 2026)
- **Efficient Reasoning & Policy Optimization** – activate, distill, and transfer knowledge under cost-aware objectives:
 - Instruction / adapter learning: [TAGI](#) (NeurIPS 2024), [SHIP](#) (ACL 2026 Main)
 - Knowledge activation & distillation: [AAG](#) (COLING 2025), [SKIntern](#) (COLING 2025), [NesyCD](#) (AAAI 2025)
 - Continual adaptation: [SpaRTA](#) (ACL 2026 Main), [PaRSP](#) (ACL 2026 Main)
 - Policy optimization: [BuPO](#) (arXiv 2025); CAPO in ResAdapt
- **Agentic Systems & Tool Learning** – reliable tool use, evaluation, and agent memory:
 - Tool learning / eval / RL: [CITI](#) (AAAI 2025), [ETAPP](#) (ACL 2025 Main), [KATE](#) (ACL 2026 Findings), [Tool-RL-Box](#) (FAGEN@ICML 2026)
 - Agent memory & preference distillation: [Distill Yourself](#)

📖 SELECTED PUBLICATIONS [[FULL LIST](#)]

(* stands for equal contribution; Listed in reverse chronological order.)

- [Huanxuan Liao](#), Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, Kang Liu. [ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning](#). (arXiv 2026)
- [Huanxuan Liao](#), Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, Kang Liu. [Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs](#). (ACL 2026 Main, CCF-A)
- [Huanxuan Liao](#), Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, Kang Liu. [Spark: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning](#). (AAAI 2026, CCF-A)
- [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Jun Zhao. [Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks](#). (AAAI 2025, CCF-A)
- [Huanxuan Liao](#), Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, Kang Liu. [SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models](#). (COLING 2025, CCF-B)
- [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Shengping Liu, Jun Zhao. [Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering](#). (COLING 2025, CCF-B)
- [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, Jun Zhao. [From Instance Training to Instruction Learning: Task Adapters Generation from Instructions](#). (NeurIPS 2024, CCF-A)
- [Huanxuan Liao](#), Shizhu He, Yupu Hao, Jun Zhao, Kang Liu. [Beyond Hard and Soft: Hybrid Context Compression for Balancing Local and Global Information Retention](#). (arXiv 2025)
- Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, [Huanxuan Liao](#), et al. [A Comprehensive Survey on Long Context Language Modeling](#). (arXiv 2025, **Core Contributor**)

👤 INTERNSHIP/PROJECT EXPERIENCE

ByteDance Beijing

Jun. 2025 – Mar. 2026

Intern Research Intern (Efficient Video Understanding)

- **Input-Side Resolution Adaptation:** Proposed **ResAdapt** with **Cost-Aware Policy Optimization (CAPO)** to allocate per-frame visual budget *before* encoding. Discards >**90%** of visual tokens while matching/exceeding strong compression baselines; under fixed compute, supports up to **16×** more frames with >**15%** relative gains on long-video reasoning (e.g., VideoMME **66.2%** on Qwen3-VL-8B + CoT at ~24% retention).
- **Compression Benchmarking & Architecture Exploration:** Reproduced multimodal visual-token compression methods; found strong random baselines competitive and temporal cues under-used. Explored efficient LLaVA variants (skip-layer, dynamic boundary compression) to widen the efficiency-accuracy Pareto frontier.

Ant Group Beijing

Feb. 2025 – May 2025

Intern Research Intern (Long Context)

- **Hybrid Context Compression:** Proposed **HyCo2**, combining soft global tokens (hybrid adapters / Q-Former-style routing) for semantics with classifier-based hard pruning for local details; matches uncompressed performance while cutting ~88.8% tokens on knowledge-intensive QA.
- **Long-Context Survey:** Core contributor to a comprehensive LCLM survey, systematizing data, architecture, and evaluation advances alongside the Awesome long-context repository.

♡ AWARDS AND SCHOLARSHIPS

Scholarships & Awards

China National Scholarship (<i>top scholarship in China; 0.2% domestically</i>), Ministry of Education	Dec. 2025
Climbing Second Prize Scholarship	Dec. 2024
Beijing Outstanding Graduate Awards, Beijing Ministry of Education	Jun. 2023
China National Scholarship (<i>top scholarship in China; 0.2% domestically</i>), Ministry of Education	Dec. 2022
China National Scholarship (<i>top scholarship in China; 0.2% domestically</i>), Ministry of Education	Dec. 2021
SiFang Society Scholarship, NCEPU	Dec. 2020

Competition

National Third Prize, Information Security Competition	Aug. 2022
National Excellence, College Student Innovation and Entrepreneurship Project	Dec. 2021
Beijing Third Prize, Internet +	Aug. 2021

💎 PROFESSIONAL SERVICES

Conference Reviewing: ACL ARR 2024–2026. ICML 2026. NeurIPS 2025. ICLR 2025. AAAI 2026.

Community: [Awesome-LLM-Long-Context-Modeling](#) (GitHub Stars > 2.2K); LCLM-Horizon survey (core contributor); [NVIDIA/kvpress](#) (contributor); [Distill Yourself](#) (agent memory & preference distillation).